26/05/2026

DISEQUAZIONI E SISTEMI DI DISEQUAZIONI DI 2° GRADO - ESERCIZI

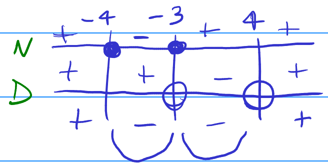
427 $\dfrac{x^2+7x+12}{x^2-x-12}\le 0$ $\quad [x\ne -3 \wedge -4\le x<4]$

N: $x^2+7x+12\ge 0 \rightarrow x^2+7x+12=0 \rightarrow x_{1,2}=\dfrac{-7\pm\sqrt{49-48}}{2}=\dfrac{-7\pm 1}{2}=\begin{cases}=-3\\=-4\end{cases}$

$x\le -4 \vee x\ge -3$

D: $x^2-x-12>0 \rightarrow x^2-x-12=0 \rightarrow x_{1,2}=\dfrac{+1\pm\sqrt{1+48}}{2}=\dfrac{1\pm 7}{2}=\begin{cases}=4\\=-3\end{cases}$

$x<-3 \vee x>4$



$\boxed{S:\ -4\le x<4 \wedge x\ne -3}$

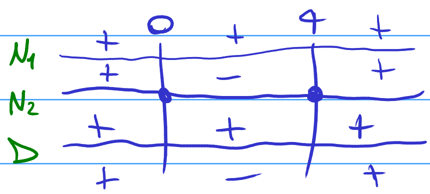
437 $\dfrac{(x^2-1)^2(x^2-4x)^3}{2x^2+1}\ge 0$ $\quad [x\le 0 \vee x=1 \vee x\ge 4]$

N: $(x^2-1)^2(x^2-4x)^3\ge 0 \rightarrow (x^2-1)^2(x^2-4x)^3=0 \rightarrow \underbrace{[(x^2-1)]^2}_{F_1}\cdot\underbrace{[x(x-4)]^3}_{F_2}=0$

F_1: $(x^2-1)^2\ge 0 \quad \forall x\in\mathbb{R}$

F_2: $x(x-4)=0 \begin{cases}x=0\\x=4\end{cases}$ $\quad x\le 0 \vee x\ge 4$

D: $2x^2+1>0 \quad \forall x\in\mathbb{R}$



S: $x\le 0 \vee x\ge 4$

533 $\begin{cases}x^3-3x+2>0\\x^6-3x^3+2>0\end{cases}$ $\quad [-2<x<1 \vee x>\sqrt[3]{2}]$

$\begin{cases}x^3-3x+2>0 & A\\x^6-3x^3+2>0 & B\end{cases}$

A $x^3-3x+2>0$

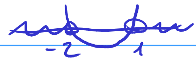
$P(1)$: $1-3+2=0$

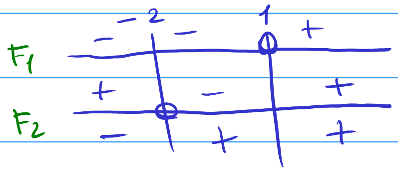
	1	0	-3	2
1		1	1	-2
	1	1	-2	0

$\underbrace{(x-1)}_{F_1}\underbrace{(x^2+x-2)}_{F_2}=0$

$F_1: \ x - 1 = 0 \rightarrow x = 1$

$F_2: \ x^2 + x - 2 = 0 \rightarrow (x+2)(x-1) = \begin{cases} x = -2 \\ x = 1 \end{cases}$





A $\boxed{-2 < x < 1 \ \lor \ x > 1}$

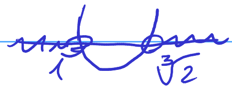
B: $x^6 - 3x^3 + 2 = 0$ $\qquad x^3 = t$

$t^2 - 3t + 2 = 0$

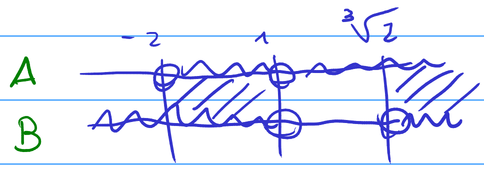
$t_{1,2} = \dfrac{+3 \pm \sqrt{9-8}}{2} = \dfrac{3 \pm 1}{2} = \begin{cases} 2 \\ 1 \end{cases}$

$t_1 = 2 \rightarrow x^3 = 2 \rightarrow x = \sqrt[3]{2}$

$t_2 = 1 \rightarrow x^3 = 1 \rightarrow x = 1$



B $\boxed{x < 1 \ \lor \ x > \sqrt[3]{2}}$



S: $-2 < x < 1 \ \lor \ x > \sqrt[3]{2}$